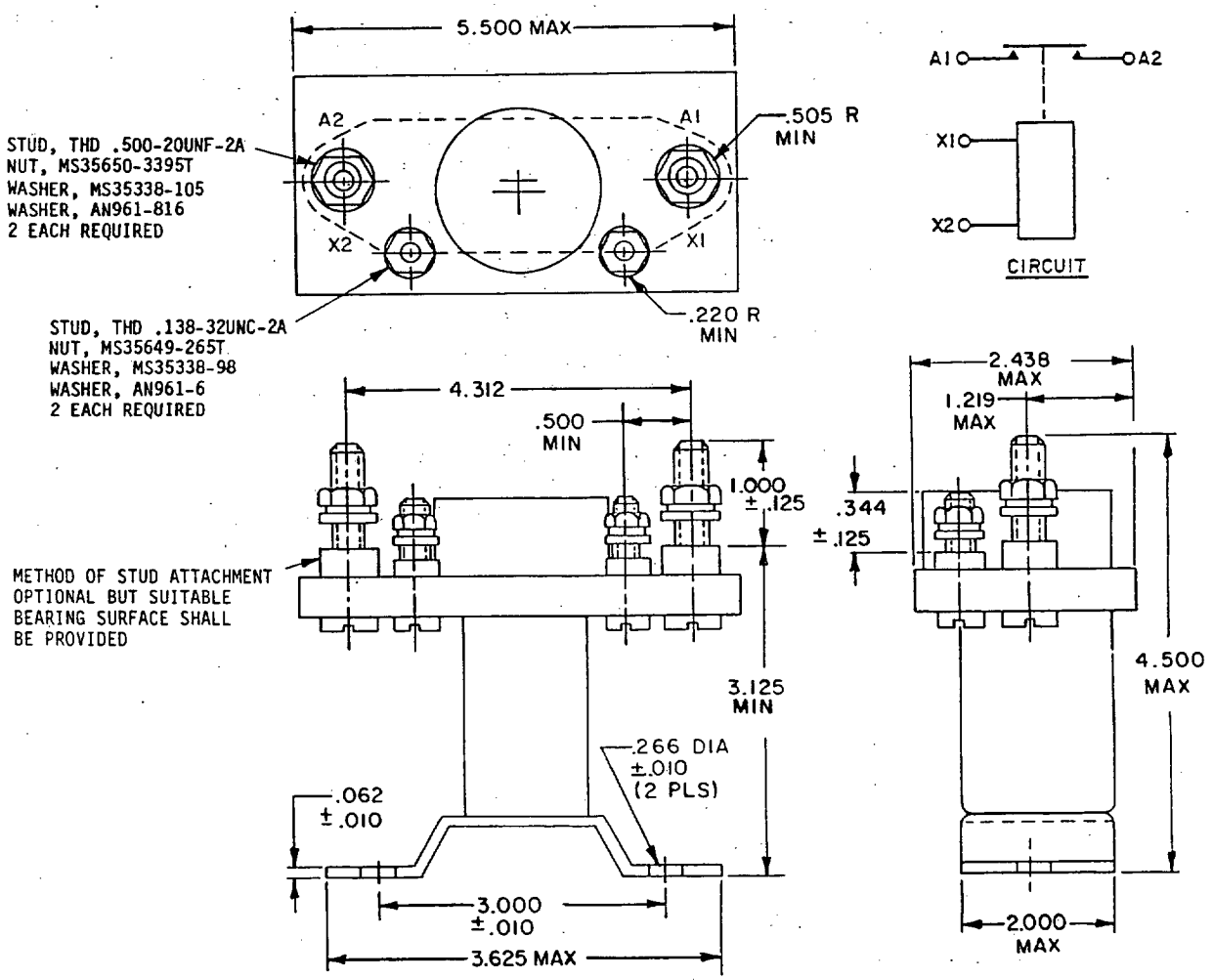
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(K) CONFIGURATION A. Outline drawing (for details see tables I and II).

(K) Denotes change

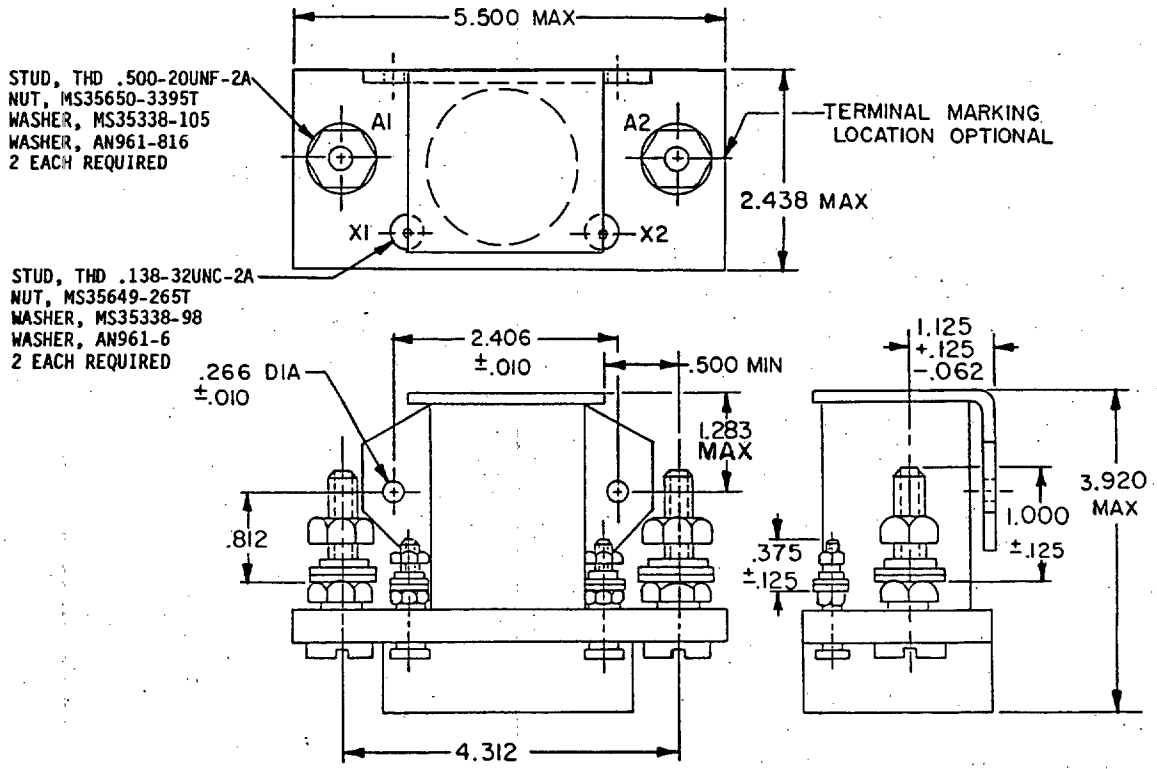
PREPARING ACTIVITY: Air Force-85 CUSTODIANS: ARMY - NAVY - AS AIR FORCE - 85 DLA - REVIEW: Air Force-99, Navy-EC USER: PROJECT NUMBER: 5945-0841-02	MILITARY SPECIFICATION SHEET TITLE RELAYS, ELECTROMAGNETIC, 400 AMPERES, 1PST (N O), TYPE III, NON-HERMETICALLY SEALED	SPECIFICATION SHEET NUMBER MS24179K 25 July 1990 SUPERSEDING MS24179J 31 Jan 1990 AMSC N/A FSC 5945
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(K) CONFIGURATION B Outline drawing (for details see tables I and II).

Inches	mm	Inches	mm
.010	0.25	1.000	25.40
.062	1.57	1.125	28.58
.125	2.36	1.219	30.96
.146	3.71	1.273	32.33
.157	3.99	2.000	50.80
.220	5.59	2.406	61.11
.266	6.76	2.438	61.92
.344	8.74	3.000	76.20
.375	9.53	3.356	85.24
.500	12.70	3.635	92.33
.505	12.83	3.920	99.57
.812	20.62	4.312	109.52
.936	23.77	4.500	114.30
		5.500	139.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. (K) Unless otherwise specified, tolerance is ±.062 (1.57 mm).
4. Additional flat washer may be used for terminal seat.
5. Terminal termperature rise under continuous current conditions, 95°C. Intermediate current to be conducted at 71°C.
6. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
7. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.
8. Shape of relay optional within envelope dimensions.

PREPARING ACTIVITY: Air Force-85 CUSTODIANS: ARMY - NAVY - AS AIR FORCE - 85 DLA - REVIEW: Air Force-99, Navy-EC USER: PROJECT NUMBER: 5945-0841-02	MILITARY SPECIFICATION SHEET TITLE RELAYS, ELECTROMAGNETIC, 400 AMPERES, 1PST (N O), TYPE III, NON-HERMETICALLY SEALED	SPECIFICATION SHEET NUMBER MS24179K SUPERSEDING MS24179J 31 Jan 1990 AMSC N/A FSC 5945
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PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY - NAVY - EC
AIR FORCE - 85 DLA -
REVIEW: Air Force-99, Navy-EC
USER:
PROJECT NUMBER: 5945-0841-02
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MILITARY SPECIFICATION SHEET

TITLE RELAYS, ELECTROMAGNETIC, 400 AMPERES, 1PST (N O), TYPE III, NON-HERMETICALLY SEALED

SPECIFICATION SHEET NUMBER: MS24179K
SUPERSEDING: MS24179J
31 Jan 1990
AMSC N/A
FSC 5945

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Ⓚ TABLE I. Dash numbers and characteristics.

Dash number MS24179-	Type	Coil	Terminal type	Mounting or mating socket	Max weight in pounds
D1 1/	III	dc	Stud	Bracket - bottom	2.6
D2 2/	III	dc	Stud	Bracket - side	2.6

1/ For Government logistics support MS24179-D1 shall be used in lieu of AN3381-2.
2/ For Government logistics support MS24179-D2 shall be used in lieu of AN3381-1.

TABLE II. Operating characteristics.

MS part no. MS24179 -	Coil data									Time (milliseconds - maximum)							
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/	Operate 3/	Release 4/	Bounce			
		Volts 1/	Freq Hz	Res Ω +15% -10%	Volts	Amperes	Normal 2/	High temp test	Cont current test					Main		Aux	
														NO	NC	NO	NC
D1	X1, X2	28	dc	9.6	29	3.4	7.5	8.75	9.4	3.0	0.5	20	15	5.0			
D2	X1, X2	28	dc	9.6	29	3.4	7.5	8.75	9.4	3.0	0.5	20	15	5.0 Ⓙ			

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

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(K) TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 v dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	400												
Inductive	10	100												
Inductive														
Motor	50	400												
Lamp														
Transfer load														2/
Mechancial life reduced current	100	100												
Intermediate minimum current														

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY - NAVY - AS; AIR FORCE - 85; DLA -
REVIEW: Air Force-99, Navy-EC
USER:
PROJECT NUMBER: 5945-0841-02
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MILITARY SPECIFICATION SHEET
TITLE: RELAYS, ELECTROMAGNETIC, 400 AMPERES, 1PST (N O), TYPE III, NON-HERMETICALLY SEALED

SPECIFICATION SHEET NUMBER: MS24179K
SUPERSEDING: MS24179J
AMSC N/A
31 Jan 1990
FSC 5945

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PREPARING ACTIVITY: Air Force-85
CUSTODIANS: ARMY - AIR FORCE - 85 NAVY-EC DLA -
REVIEW: Air Force-99, Navy-EC
USER:
PROJECT NUMBER: 5945-0841-02
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MILITARY SPECIFICATION SHEET
TITLE: RELAYS, ELECTROMAGNETIC, 400 AMPERES, IPST (NO), TYPE III, NON-HERMETICALLY SEALED

SPECIFICATION SHEET NUMBER: MS24179K
SUPERSEDING: MS24179J
31 Jan 1990

AMSC N/A
FSC 5945

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ENVIRONMENTAL CHARACTERISTICS

Temperature range	-55° to +71°C
Maximum altitude rating	50,000 ft
Shock G-level	25 g's
Duration	6-9 ms
Max duration contact opening	2 ms
Vibration - Sinusoidal (see table IV)	
Vibration - Random	N/A
High shock	N/A
Acceleration	10 g's

TABLE IV. Vibration levels.

5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1500 Hz
.08	.06 DA	2 g's	2 g's	

Qualification by similarity (see table V)

TABLE V. Qualification by similarity.

Part number MS24179	Loads						Dynamics 1/			Environmental		
	Type 1			Type 1 ER								
	A	B	C	D	E	F	A	B	C	A	B	C
D1	4						2			4		
D2	4						2			4		

1/ For Group C testing, each dash number shall be tested to meet vibration requirements.

ELECTRICAL CHARACTERISTICS

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms
Dielectric strength (sea level)	2-5 seconds

	Inital		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250 V rms	N/A	1,000 V rms	N/A
Aux contacts	1,250 V rms	N/A	1,000 V rms	N/A
All other points	1,800 V rms	N/A	1,350 V rms	N/A

Dielectric strength (altitude) 1 minute

	28 V dc	115 V ac
Coil to case	500 V rms	N/A
Aux contacts	500 V rms	N/A
All other points	500 V rms	N/A

Max contact drop initial	0.175 volt
After life test	0.200 volt
Overload current	(NO) 3200 amperes
Rupture current	(NO) 4000 amperes
Duty rating	Intermittent
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

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